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Quantitative Model to ETF and Portfolio Analysis

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Exchange-Traded Funds (ETFs) have reshaped modern finance by presenting a complex but rewarding system that allows for more diversification and ease to trade. This work proposes a systematic quantitative framework for ETF evaluation, portfolio construction, and risk management. Using 2020–2025 data, it builds a deterministic screening and scoring system across three dimensions: replication accuracy, cost efficiency, and liquidity. It then integrates eligible funds into portfolios under active allocation strategies, comparing Naive Risk Parity and Volatility Targeting, achieving a value of β of approximately 1.0 and 0.5, respectively. In order to maximize our return, we also automated Calendar and Drift rebalancing, obtaining an annual return of 8% and 10%, respectively. Finally, the study develops a forward-looking risk assessment using Monte Carlo simulations with Cholesky decomposition within a Geometric Brownian Motion (GBM) setting. We verify that this approach successfully preserves asset co-movements, capturing diversification and central performance tendencies. Nevertheless, we also note the inability to predict excessive shocks on the market.

I. Introduction

Trade systems have served the fundamental foundation of economics growing to become a core pillar of civilization, often operating as a structured system of shared beliefs.

In 1952, a groundbreaking and Nobel Prize-winning framework was introduced: Modern Portfolio Theory (MPT), developed by Harry Markowitz (Ref. [1]). MPT elegantly combines Mathematics, Probability, and Statistics with Economic theory, assuming that investors operate under a coherent system of probability beliefs to present an optimized method for portfolio construction. It also mentioned the importance of risk diversification in a portfolio.

A portfolio is a structured allocation of financial assets in specific proportions. This allocations are driven to change in accord to the market movements. The most simple case of a portfolio is the Buy and Hold (BH), where the investor just defines the initial allocation of each asset and then does not make any more changes.

The asset wher we will be placing greater focus are the investment funds which allow for a greater diversification. They are professional managed, pooling capital from multiple investors to invest across a wide range of financial assets. They first appeared in Europe during the early 1800s (Ref. [2]).

The funds in study will be Exchange-Traded Funds (ETFs), which offer the diversification of traditional funds and allow an efficient trade. ETFs reply an index fund - market barometers which make of the market measurable. This way, they are low in cost, since they do not need asset picking.

However, there is little consensus on the creation of ETFs portfolios. This proposal aims to facilitate the evaluation and selection of ETFs based on their stand-alone performance and their comparison relative to a benchmark index. Additionally, it aims to simulate and provide quantitative analysis of investment portfolios constructed specifically from these ETFs.

Because forecasting asset prices is central to maximizing investment outcomes, various mathematical frameworks have been created to predict market behaviour. One of the most widely accepted models, introduced by Paul Samuelson, demonstrates how Geometric Brownian Motion (GBM) can effectively model stock price trajectories(Ref. [3]). This paper extends that definition to forecast overall portfolio prices. By incorporating a Cholesky decomposition to capture how distinct assets co-move over time, the proposed model implements these asset paths using Monte Carlo simulations.

II. ETF assessment

The first stage of the proposed system is a quantitative and deterministic procedure to select and evaluate ETFs. Each candidate ETF is characterised along three dimensions: replication, cost and liquidity.

A. Replication metrics

Correlation

We make use of The Pearson correlation to compare the ETF and its benchmark. It takes values in $[-1, 1]$. Values close to 1 indicate that the ETF closely follows its benchmark (negative values mean that they are inversely correlated and move in opposite directions).

$$\rho_{\text{etf,bm}} = \frac{\text{Cov}(r_{\text{etf}}, r_{\text{bm}})}{\sigma_{\text{etf}}\,\sigma_{\text{bm}}} \tag{1}$$

where r_{etf} and r_{bm} denote the ETF and its benchmark returns, respectively, σ_{etf} and σ_{bm} are the standard deviations of the respective returns.

Beta (β)

Beta measures the sensitivity of the ETF returns to the movements of its benchmark:

$$\beta_{\text{etf}} = \frac{\text{Cov}(r_{\text{etf}}, r_{\text{bm}})}{\sigma^2_{\text{bm}}} \tag{2}$$

A $\beta > 1$ indicates that the ETF is more volatile than its benchmark and a $\beta < 1$ that it is less volatile. For a fund whose goal is to replicate an index, ideally, we should have $\beta \approx 1$. β can also be explained by the slope of the linear regression:

$$r_{\text{etf},t} = \alpha + \beta\, r_{\text{bm},t} + \varepsilon_t \tag{3}$$

whose coefficient of determination R^2 reports the fraction of the ETF return variation that is explained by the benchmark. Since the explanatory variable is the benchmark itself, $R^2 = \rho^2_{\text{etf,bm}}$. The pair (β, R^2) is more informative than the beta alone as, for example, a $\beta \approx 1$ accompanied by a high R^2 confirms that the beta genuinely reflects index tracking.

Tracking Error (TE)

Is the standard deviation of the difference between the ETF and benchmark returns, $d_t = r_{\text{etf},t} - r_{\text{bm},t}$:

$$TE = \sqrt{\frac{\sum_{t=1}^{T}(d_t - \bar{d})^2}{T-1}} \tag{4}$$

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The smaller the tracking error, the closer the ETF returns are to those of its benchmark. This metric aims to capture the magnitude of the deviations not distinguishing between favourable and unfavourable deviations.

B. Cost metrics

Total expense ratio (TER)

TER is the annual percentage charged to investors holding units of the fund, arising from its management, maintenance and operating costs. It reduces the net return delivered to the investor, so lower values are preferred.

C. Liquidity metrics

Liquidity refers to the ease of buying and selling units of the fund without materially affecting its price. It is represented by:

Average Daily Traded Value (ADTV)

The traded value on a given day is the number of units traded multiplied by their price, so it expresses the economic footprint. ADTV is the average of this quantity over the estimation window of W days:

$$\text{ADTV} = \frac{1}{W}\sum_{t=1}^{W} p_t\, v_t \tag{5}$$

where p_t is the close price and v_t the traded volume at day t. The higher the ADTV, the larger positions can be traded without significantly moving the market.

Bid-Offer Spread

The spread is the difference between the lowest price a seller will accept (p_{ask}) and the highest price a buyer will pay (p_{bid}):

$$\text{spread} = p_{\text{ask}} - p_{\text{bid}} \tag{6}$$

A wider spread indicates lower liquidity. We will represent this as a percentage of the ask price, which represents the minimum appreciation the asset must achieve to reach *breakeven*:

$$\text{spread}(\%) = \frac{p_{\text{ask}} - p_{\text{bid}}}{p_{\text{ask}}} \times 100 \tag{7}$$

D. Eligibility filtering

Before scoring, we need to define our acceptability thresholds. Any fund that violates at least one limit is flagged as ineligible and excluded, which can be seen in Table XII(Appendix A).

The minimum correlation of 0.95 comes from the passive-management criterion that the this relationship, $R^2 \geq 0.9$ Ref. ([4]), is satisfied. This implies that $\rho \gtrsim 0.95$. β is bounded on both sides because a faithful replication should react exactly as its index. TER and spread carry maximum limits because lower values are preferable, while ADTV carries a minimum because higher traded value indicates greater liquidity.

E. Scoring system

To compare metrics, each eligible metric is mapped onto the interval $[0, 1]$ through a min-max normalisation, where 1 denotes the best possible outcome for that metric. For that, we can have three types of functions: monotone increasing, monotone decreasing and target value (Appendix B). Correlation and ADTV are scored as monotone increasing; TER and spread as monotone decreasing; β as a target metric centred on 1.

The metric scores are then combined into three category scores (replication, cost and liquidity) through a weighted average within each group. The category scores are combined into a single total score through a weighted average across groups. The weights used in this work are reported in Table I. The weighting distribution was a suggestion of the supervising company.

TABLE I: Scoring weights within and across metric groups.

Group (group weight)	Metric	Metric weight
Replication (0.34)	Correlation	0.3
	Beta	0.4
	TE	0.3
Cost (0.33)	TER	1.0
Liquidity (0.33)	ADTV	0.6
	Spread	0.4

III. Portfolio Rebalancing Methods

The following proposed strategies have a common ground: they rely heavily on the concept of rebalancing, the act of recapturing the original risk characteristics of the portfolio, correcting asset allocations so that they correspond to the initially defined ones.

However, it is necessary to keep in mind that every transaction has a cost. Therefore, it is mandatory to define adequate conditions which can trigger rebalancing. Ref. [5] concludes that, for equities and bonds funds based portfolios, the impositions of the following conditions is likely to produce an appropriate trade-off between the level of risk taken and the reduction of costs: (1) annual or semi-annual monitoring, (2) rebalancing at 5% thresholds. This leads us to the definition of calendar rebalancing and of drift rebalancing, respectively, experimenting more conditions.

A. Calendar Rebalancing

This method consists on reallocating the assets within the portfolio at pre-determined interval times, so it allows them to agree with the target weights, every time rebalancing is triggered.

At the end of each rebalancing period, it compares the current weight allocation of the ith asset, $w_{i,t}$, within the portfolio with the target w_i^T. If:

1. $w_{i,t} < w_i^T$: an order to buy units of asset i will be placed, so it increases its allocation within the portfolio
2. $w_{i,t} > w_i^T$: ian order to sell units of asset i will be placed, so it decreases its allocation within the portfolio.
3. $w_{i,t} = w_i^T$: asset i does not suffer any changes in its allocation.

B. Drift Rebalancing

This approach studies the drift of the assets allocations within the portfolio with the goal to reallocating the asset weights in agreement with the target ones. However, there is a twist: it attributes a deviation tolerance band from the target weight of the asset. Therefore, there is a space for the asset to drift. The proposed method triggers rebalancing if at least one of the assets drifts further than the allowed. This way, when there is a need for rebalancing, even if it was just one asset, every asset will be forced to meet its target positions.

Step 1: Daily monitoring of the asset weight drift ($w_{i,t}$) within the portfolio.

Step 2: Compute the deviation $d_{i,t}$ of the current i^{th} asset weight from its target w_i^T - this is the drift:

$$d_{i,t} = w_{i,t} - w_i^T \tag{8}$$

Step 3: Assess the deviation from the tolerance band. If the following condition is met, then it executes the 4th step:

$$\exists i \in \{1, 2, ..., n\} : |d_{i,t}| > \delta. \tag{9}$$

If not, then the algorithm just proceeds to continue the BH method.

Step 4: The necessary units are traded so that the current weights can meet the target weights.

IV. Portfolio Construction Methods

A. Risk Parity

The Risk Parity (RP) approach was born from the need for diversification of portfolio risk exposure, which was lacking in

traditional portfolios, as noted in Ref. [6], as well as the need to diverge from the unreliability of expected returns in the construction of optimal portfolios (Ref. [7]).

RP portfolios rely solely on risk-based diversification, aiming to prevent a single asset or a asset class from dominating the portfolio's risk exposure. To achieve this, a specific weight is assigned to each asset within the portfolio, such that it contributes equally to the portfolio total risk budget [8].

1. Naive Risk Parity (NRP)- Method

NRP relies solely on *volatility* as the risk measure. The process to achieve the perfect weight allocation is:

Step 1: Compute assets volatility until day t:

$$\sigma_{i,t} = \sqrt{\frac{\sum_{i=1}^{n}(r_{i,t} - \overline{r_i})^2}{T-1}} \tag{10}$$

where $r_{i,t}$ corresponds to the return of asset i at time t.

Step 2: Compute assets inverse volatility: $\frac{1}{\sigma_{i,t}}$

Step 3: The *target weights* to be achieved in day $t+1$ are normalized using the following formula:

$$w_i = \frac{\frac{1}{\sigma_{i,t}}}{\Sigma_{j=1}^{N}\frac{1}{\sigma_{j,t}}} \tag{11}$$

This approach ensures that (1) there is no look-ahead bias when computing asset volatility as it is only used information that is known until day t and the order is executed in day t+1 (2) the weights are inversely proportional to the volatility, meaning that the asset that carries more risk has a lower allocation.

B. Volatility Targeting (VT)

This approach implies that the portfolio is managed such that its risk exposure is in agreement with a target volatility.

1. Method

The target idea of this method is to leverage the risk exposure of the portfolio in accordance with the relationship between the current volatility and the target one.

Our proposed method takes advantage of the approach of [9] but adapts the strategy for it to be possible to evaluate portfolios and not just assets.

Step 1: The target volatility, σ^T for each day is set to be the a rolling window of the realized volatility of the benchmark portfolio, stopping at the evaluation day.

Step 2: At the assessment day, t, the adjustment to the portfolio's risk exposure is computed making use of the leverage:

$$L = \frac{\sigma^T}{\sigma_t} \tag{12}$$

where σ_t represents the computed volatility using information available until day t to avoid look-ahead bias.

Step 3: Still in the assessment day, the allocation of each asset on the portfolio is then multiplied to the computed leverage, so it can gives us the *target weights*.

So, following this logic, what happens to the risk exposure of the portfolio can be translated by:

1. $\sigma^T \leq \sigma_t \Rightarrow L \leq 1$: the assets allocation within the portfolio is reduced, thus the exposition of the portfolio to risk is reduced, lowering the overall volatility of the portfolio.
2. $\sigma^T \geq \sigma_t \Rightarrow L \geq 1$: the assets allocation within the portfolio is increased, thus the exposition of the portfolio to risk is increase, increasing the overall volatility of the portfolio.

V. Portfolio Assessment

To understand the portfolio performance it is necessary to compute certain metrics. The metrics can be divided into two categories: Ex-Ante or Ex-Post. The first one refers to the forecast of the performance of the portfolio, meaning it tries to predict the portfolio's behaviour before the evaluated period. The second one evaluates the exact performance of the historical returns of the portfolio (Ref. [10]).

A. Ex-Post

Ex-post risk measures are very intuitive because they take advantage of the realized portfolio returns, hence they provide an accurate measurement of past metrics. Despite that, they can fail as a good predictor for the future, since future portfolio's natural evolution is not accounted (Ref. [11].)

Returns

The first step towards the understanding of portfolio realized performance is computing the realized returns $R_{p,t}^{\text{ex-post}}$:

$$R_{p,t}^{\text{ex-post}} = \sum_{i=1}^{n} w_{i,t-1} \times r_{i,t}, \tag{13}$$

where $w_{i,t-1}$ is the latest allocations of the asset on the portfolio and and $r_{i,t}$ are the returns of each individual asset. Adding each parcel gives the full portfolio return for day t.

Volatility

Portfolio volatility is the standard deviation of the returns of the portfolio. It measures the magnitude of its returns fluctuations over a period of time. It differs from asset volatility because it needs to take into account the correlation between the portfolio's assets.

$$\sigma_{p,t}^{\text{ex post}} = \sqrt{\frac{\sum_{i=1}^{n}(R_{p,t} - \overline{R_p})^2}{T-1}} \tag{14}$$

Excess Returns

Excess returns are used to understand the outperformance or underperformance of an investment relative to its benchmark.

$$ER_p = R_p^{\text{ex post}} - R_{bm}^{\text{ex post}} \tag{15}$$

Maximum Drawdown (MDd)

MDd is defined as the largest cumulative decline from a historical peak in portfolio value to the subsequent trough. It quantifies the extent to which losses can be prolonged. Significant drawdowns are often associated with increased investor redemptions and liquidity pressure, making this metric a critical indicator for assessing the risk-adjusted performance.

Alpha (α) and Beta (β)

As we have already seen, there is a linear regression used to assess β, which will also provide a value for α (Eq. 3). This formula is adapted to:

$$R_{p,t}^{\text{ex-post}} - R_{f,t}^{\text{ex-post}} = \alpha + \beta(R_{bm,t}^{\text{ex-post}} - R_{f,t}^{\text{ex-post}}) + \varepsilon_t, \tag{16}$$

where $R_{f,t}^{\text{ex-post}}$ is the underlying risk free return portfolio. Hence, β will allow to understand if the portfolio is outperforming or under-following the benchmark chosen and α will give information about the reason of the deviation. α measures a strategy's ability to earn returns independent of market direction. A positive α shows risk-adjusted outperformance net of systematic risk, while a negative α suggests a drag from costs or poor timing. β measures the strategy's sensitivity to its benchmark: values above 1.0 magnify market moves (and thus volatility and drawdowns), values between 0 and 1.0 capture only part of market risk.

Information Ratio (IR)

The IR quantifies the excess return that comes from taking additional risk using a active management strategy. If it turns out negative it means that the investment was not worthy.

$$IR = \frac{\overline{ER_p}}{\sigma_{ER_p}} \tag{17}$$

B. Ex-Ante

Ex-Ante measures can be a great tool to try to predict the portfolio's future performance making use of the current holdings of

the portfolio. However, as they are based in historical statistics of asset returns, they are not necessarily a good predictor of future behaviour. Nevertheless, they have the advantage of not needing historical information about the portfolio, only current market allocations and assets' expected returns.

Expected Returns

These are forecasts and can be either computed using prediction Stochastic models, Geometric Brownian Motion (GBM) dynamics and Monte Carlo methods or by a simple covariance matrix until $t-1$ of the historical asset returns and future allocations within the portfolio. As a consequence, we present a method that uses a simple estimation of the covariance of the asset returns and further in the paper also a Monte Carlo model that combines GBM with Stochastic Calculus:

$$E[R_{p,t}] = \sum_{i=1}^{N} w_{i,t} \cdot \mu_{i,t-1}, \tag{18}$$

where $w_{i,t}$ it is the allocation weight of asset i at time t and $\mu_{i,t-1}$ is the expected return of asset i for time t, calculated as the rolling 21-day mean return lagged by one period (t-1) to strictly eliminate look-ahead bias.

Volatility

Ex-ante volatility refers to a forward-looking assessment of a portfolio's volatility (Ref. [10]).

$$\sigma_{p,t}^{\text{ex ante}} = \sqrt{\frac{\sum_{i=1}^{n}(R_{p,t}^{\text{ex ante}} - \overline{R_p^{\text{ex ante}}})^2}{T-1}} \tag{19}$$

Value at Risk (VaR)

VaR is a risk metric that associates the magnitude of a potential loss in a given portfolio with the probability of its occurrence. (Ref [12]).

Its computation relies on modelling the probability distribution of a portfolio's potential gains and losses over a defined holding period. By isolating the downside tail of this distribution relative to a chosen confidence level, the specific VaR threshold is established. It is important to notice that this approach may result in neglecting the potential extreme losses.

Choosing a confidence level of 95% means that the probability of the potential loss surpass the figure that appears is 5%:

$$P(Z < z_{0.95}) = 0.95. \implies z_{0.95} = 1.645 : VaR = \sigma_{p,t}^{\text{ex post}} \times 1.645 \tag{20}$$

Conditional Value at Risk (CVaR)

To take into account the extreme losses that VaR overlooked, CVaR is computed. It measures the expected value of the extreme losses, meaning that it measures the expected loss in the tail of a distribution beyond a certain quantile, i.e. 95%. Thus, CVar provides an accurate characterization of the risk, particularly as manifested in the extreme tails of the distribution.

$$\text{CVaR}_p = E\left[L \mid L > \text{VaR}_p\right] \tag{21}$$

where L denotes the loss as a random variable (i.e. the possible value of actual loss).

C. Geometric Brownian Motion, Cholesky decomposition and Monte-Carlo approach

To simulate the historical or forward-looking evolution of the portfolio, we utilize a continuous-time stochastic process framework. A stochastic process is defined as a parametrized collection of random variables $\{S(t)\}_{t\in T}$ indexed by time, characterizing the dynamical evolution of a system under uncertainty (Ref. [13]).

Under this framework, asset price dynamics are assumed to follow a GBM, governed by the stochastic differential equation (SDE):

$$dS(t) = \mu S(t)\,dt + \sigma S(t)\,dW(t) \tag{22}$$

where $S(t)$ denotes the asset price at time t, μ is the constant drift parameter representing the expected rate of return, σ is the volatility coefficient, and $dW(t)$ represents the infinitesimal increments of a standard Wiener process (Brownian motion).

By applying Itô's Lemma to the non-linear transformation $f(S) = \ln S(t)$, the SDE can be integrated analytically (see Appendix F for the full derivation). This yields the exact solution for the asset price at a future time horizon $\tau > t$:

$$S(\tau) = S(t)\exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)(\tau - t) + \sigma\left(W(\tau) - W(t)\right)\right] \tag{23}$$

where $(\tau - t)$ defines the forecast horizon. Within this simulation framework, the parameters μ and σ are treated as constant historical estimators, mapped directly to the portfolio's annualized mean returns and realized volatility computed up to time $t-1$, respectively.

In order to faithfully simulate a portfolio using different assets, we have to also take into account the possible correlations that exist between them. For that, we follow the approach presented in Ref. [14]. As a result, we are going to start by recalling a very powerful linear algebra theorem and tool:

Theorem (Cholesky Decomposition): Let $B \in \mathbb{R}^{n\times n}$, a positive-definite, real and symmetric matrix. There exists and unique $L \in \mathbb{R}^{n\times n}$ lower-triangular matrix with positive diagonal, s.t.

$$B = LL^T \tag{24}$$

Let $S_1(t)$ and $S_2(t)$ be the asset price dynamics given by the following stochastic differential equations:

$$dS_1(t) = \mu_1 S_1(t)\,dt + \sigma_1 S_1(t)\,dW(t)_1, \tag{25}$$

$$dS_2(t) = \mu_2 S_2(t)\,dt + \sigma_2 S_2(t)\,dW(t)_2, \tag{26}$$

where $dW(t)_1$ and $dW(t)_2$ are the Wiener terms with correlation $\text{Cov}(dW(t)_1, dW(t)_2) = \rho\,dt$. We then define the covariance matrix

$$\Delta = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \tag{27}$$

These coefficients can be computed using historical data. We can then perfomr the Cholesky decomposition on $\Delta = LL^T$, where

$$L = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix} \tag{28}$$

which then becomes

$$LL^T = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix}. \tag{29}$$

From here, it is trivial to obtain the equations

$$l_{11} = \sigma_1, \quad l_{21} = \rho\sigma_2, \quad l_{22} = \sigma_2\sqrt{1-\rho^2}. \tag{30}$$

In order to simulate, using a Monte-Carlo analysis, we start by sampling ϕ_1 and ϕ_2 from $\mathcal{N}(0,1)$ independently. Then, we compute $\tilde{W}_1$ and $\tilde{W}_2$, using matrix L,

$$\begin{pmatrix} \tilde{W}_1 \\ \tilde{W}_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ \rho\lambda_2 & \lambda_2\sqrt{1-\rho^2} \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \tag{31}$$

This transformation ensures that the new variables $\tilde{W}_1$ and $\tilde{W}_2$ reflect the desired correlation structure, facilitating accurate simulation of the dynamics of the underlying assets in a complex financial market setting.

After applying the change of variables using the Cholesky decomposition, the new variables $\tilde{W}_1$ and $\tilde{W}_2$ are used to simulate

the asset price paths for S_1 and S_2. For each time step Δt, the updated asset prices are given by

$$S_1(t+\Delta t) = S_1(t)\exp\left[\left(\alpha_1 - \tfrac{1}{2}\sigma_1^2\right)\Delta t + \sigma_1\sqrt{\Delta t}\,\tilde{W}_1\right], \quad (32)$$

$$S_2(t+\Delta t) = S_2(t)\exp\left[\left(\alpha_2 - \tfrac{1}{2}\sigma_2^2\right)\Delta t + \sigma_2\sqrt{\Delta t}\,\tilde{W}_2\right] \quad (33)$$

This was a case where we have 2 assets. However, using the Cholesky decomposition, the process for n assets is completely analogous and trivially scalable.

VI. Applications

A. Portfolio Construction

The proposed work aimed to build a method that could analyse the behaviour of the proposed portfolios when applied with the previous 2 construction strategies. However, to turn it in an even more useful system, it is possible to apply the rebalancing methods. This way, we have designed a system that does not just allow the construction of a one time portfolio but also allows its continuous rebalancing, when the trigger conditions are met.

The architecture of the underlying algorithm is then structured as the following procedure.

1. *Initial units and Initial invested value*

Step 1: The process of the physical creation of a portfolio starts with the purchase of the units corresponding to the initial desired allocations of the assets in the portfolio. The initial units for asset number i are computed as:

$$u_{i,0} = \frac{V_{p,0} \times w_{i,0}}{v_{i,0}} \quad (34)$$

where $V_{p,0}$ is the total desired value to invest on the portfolio, $w_{i,0}$ is the initial desired allocations (weights) of asset i on the portfolio and $v_{i,0}$ is the price of the asset at purchase time.

Step 2: As discussed, one can just estimate the behaviour of the asset at the opening of the markets. And there is also the need to account for transaction costs. So, the first step towards the understanding of the value that is necessary to inject to get the desired capital is to use the close prices of asset i of the day previous to the purchase $(t-1)$ $v_{i,close(t-1)}$ on the denominator of Eq. 34. We now have the prediction of the number of units we are going to buy.

Step 3: The estimated value of the investment should then be:

$$\hat{IV}_p = \left(\sum_{i=1}^{n} u_{i,0} \times v_{i,t-1(close)}\right) \quad (35)$$

Step 4: Then, one should account for the total value of transactions costs, such that its prediction is given by:

$$\hat{TC} = \hat{IV}_p \times \left(\frac{\text{bps}}{10000}\right) \quad (36)$$

where bps stands for basis points and represents the measure of transaction costs.

Step 5: So the estimated necessary capital to inject is formulated as:

$$\hat{IC} = \hat{IV}_p + \hat{TC} \quad (37)$$

Step 6: On the next day, the real units are purchased. Due to the gap between close and opening prices, the true value of the investment is known by replacing $v_{i,0}$ with $v_{i,open(t)}$ in Eqs. 34 and 35. The real values for Eqs. 35, 36 and 37 follow from this substitution.

2. *Buy & Hold until Trigger condition*

According to each one of the methods proposed, the algorithm continuously detects if the trigger condition is activated. This conditions can be found on Appendix C.

If these trigger conditions are not met, the portfolio just proceeds with BH method until it triggers rebalance. However, if the trigger is activated, the algorithm advances into the next stage.

3. *Rebalancing - Evaluation - day t*

At the end of day t, a comprehensive evaluation is conducted to identify the mandatory adjustments required to ensure that the asset allocations within the portfolio comply with the *target weights*:

Method	Target weight Computation
NRP	By the inverse volatility: Eq. 11
VT	With target risk exposure and leverage: Eq. 12
CR	Equal to the initial weights
DR	Equal to the initial weights

TABLE II: Computation of target weights

Step 1: Compute the NAV at close:

$$NAV_c = \left(\sum_{i=1}^{n} u_{i,t} \cdot v_{i,t(close)}\right) + \text{ cash} \quad (38)$$

where $u_{i,t}$ is the i-th asset units at day t, that were acquired the last time, and $v_{i,t(close)}$ is the i-th asset close price on day t.

Step 2: Estimate target values for each asset:

$$\hat{TV}_{i,t} = NAV_c \times w_{i,t}^T, \quad (39)$$

where w_t^T is the target(T) weight for asset i at time t.

Step 3: Compute the target units to get the desired weight allocation in the portfolio:

$$u_{i,t}^T = \frac{\hat{TV}_{i,t}}{v_{i,t(close)}} \quad (40)$$

Step 4: Discover the units that are going to be traded (u_i^{Td}):

$$u_i^{Td} = u_{i,t}^T - u_{i,t} \quad (41)$$

If $u_i^{Td} < 0$: the number of units is greater than the need, thus the difference is going to be sell.
If $u_i^{Td} > 0$ the number of units is lesser than the need, thus the difference is going to be bought.

4. *Rebalancing - prediction of costs - day t*

To reduce as much as we can the need for capital injection, we follow the following steps.

Step 1: Compute the cash flow from sells and buys:

$$Cash^a = \sum_{i=1}^{n} u_i^a \times v_{i,t(close)}^a, \quad (42)$$

where a denotes the action: sell or buy.

Step 2: Compute estimated transaction costs:

$$\hat{TC}_t = \left(\sum_{i=1}^{n} \hat{TV}_i\right) \times \frac{bps}{10000} \quad (43)$$

Step 3: Determine whether a capital injection is required. In this section, the goal is to minimize as much as possible the need for additional capital. Therefore, the algorithm first simulates the execution of the sell orders, followed by the simulation of the buy orders. If the proceeds from the sells are insufficient to cover the cost of the buys and all the associated transaction fees, the shortfall must be covered through a capital injection.

5. *Rebalancing - Execution and true costs at day t + 1*

In this section, steps 3 and 4 from the Evaluation stage (Section VI A 3) and also steps 1, 2 and 3 from the prediction stage (Section VI A 4 are executed but assuring that the asset prices that are used to make the calculations are the opening prices at $t+1$, $v_{i,t+1(open)}$. This way, we know for sure the true costs of the transaction.

Note that a cash account is initialised with the initial investment on the beginning of the program. The cash flow is updated whenever it is necessary. In every strategy, if there is the need to purchase more units of the assets, when it's time for the execution, all the cash available in the portfolio is reinvested in assets. This way the total value of injection of capital can be reduced. This account allows for negative balance. Only this way we can discount the liabilities (capital injected to cover shortfalls) from the NAV to get true cumulative returns.

VII. Results and Analysis

A. ETF assessment

In this section, we report the results of the ETFs that we focused our analysis on.

ETF	BM	β	R^2	ρ	TE(%)	TER(%)	ADTV (€)	Spread (%)
D	H	0.995	0.997	0.999	0.5	0.12	5,747,350	0.0424
B	F	0.871	0.907	0.952	0.12	0.08	1,788,125	0.0563
C	G	1.019	0.996	0.998	0.06	0.21	46,313,760	0.0124
A	E	1.029	0.991	0.995	0.14	0.07	4,163,900	0.0326

TABLE III: ETF assessment metrics relative to each tracked index.

As we can observe on Table III, all the ETFs were eligible, as all of them are inside the bonds that were proposed on Table XII. Not only that, for all assets, we check that for β, R^2 and ρ its values are very close to 1. This means that the ETF follows closely the benchmark it is supposed to reproduce. This is also confirmed by the low percentage on the TE. Additionally, the value on TER means that the annual percentage charged to the investors is rather low. Finally, the values on ADTV confirm their high liquidity.

Class	ETF	Repl.	Cost	Liq.	Total	Rank
Equity	D	1.000	1.000	1.000	1.000	1
	C	0.980	1.000	1.000	0.993	2
	A	0.948	0.721	1.000	0.890	3
Bonds	B	0.460	0.950	1.000	0.800	1

TABLE IV: Final ETF scoring per asset class. Scores range from 0 to 1, with 1 being the best within the criterion. All listed ETFs passed the eligibility filters.

By observing Table IV, we check that all the assets perform very close to each other. As we can see, D seems to perform the best with C performing very close to it, as we can confirm by the all the values very close to 1. A lags a bit behind the previous two, especially on the Cost, presenting a good performance overall, nonetheless.

Lastly, in table III, we check that B performs baddly according to the metrics defined to Replication. However, this can be very simple explained: the problem is that not just B, but also C and D are being compared to other ETFs that follow the respective indexes, since there was not available free data of this indexes. However, its interesting to notice that B and the respective comparison ETF do not have the same performance, although they are supposed to follow the same index (this decision was confirmed by the supervising company).

B. Portfolio Construction

The analysis will be focus on the comparison between different rebalancing dates in the different rebalancing approaches, and then on the comparison of the performance of the rebalancing concept applied to the methods of portfolio construction (risk parity and volatility target). It will also focus on evaluating how the different strategy-applied portfolios perform relatively to the same portfolio using BH method.

The portfolio was constructed using 3 different ETFs, such that their initial allocation was:

ETF	Allocation	Class
A	0.2	Bonds
B	0.2	Equity
C	0.6	Equity

TABLE V: Initial Allocations and Class

The data was collected between 01-01-2020 and 30-12-2025 and the initial investment is 9995€ (the TC were 5€ so the total amount was 10000€). The assessment figures of BH performance, used to make the comparison in the following analysis can be seen in Appendix D.

C. Calendar Rebalancing

The rebalancing periods (in days) chosen to compare were 21 (monthly rebalancing), 60 (quarterly), 126 (semi-annual rebalancing) and 252 (annual rebalancing). The plot refers to the total cumulative return of the investment.

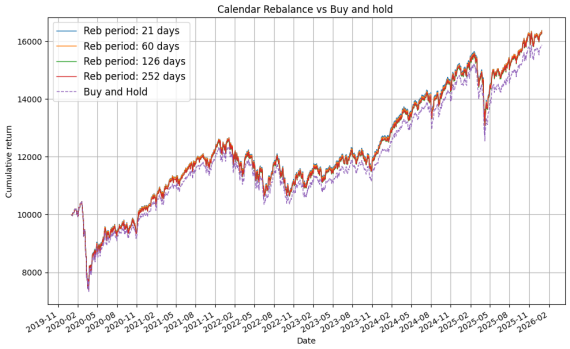


FIG. 1: Cumulative Ret. for diverse rebalancing periods - CR

To compare the returns of each:

Reb.	Inv. (€)	Final (€)	Annual (%)	Abs. Ret. (%)	Exc. Ret. (%)
21	9,995.00	16,371.53	8.58	63.80	+0.56
60	9,995.00	16,363.75	8.57	63.72	+0.55
126	9,995.00	16,299.37	8.50	63.08	+0.48
252	9,995.00	16,313,17	8.51	63.21	+0.49

TABLE VI: Calendar Rebalancing — Comparison of strategies

Analysing Fig. 1 and table VI:

- Rebalancing Advantage: the various rebalanced portfolios achieve higher cumulative returns when compared to the BH approach with static units holdings. This suggests that continuously capturing market natural evolutions through periodic optimization can translate into enhanced performance, if the market is in fact providing that growth.
- Rebalancing Frequency: the results that were achieved denote that higher cumulative returns are in decrescent order: 21, 60, 252, 126 and never (BH).
- Capital Injection: cumulative returns did not take into account capital injection so it could capture the true performance of rebalancing. However, having this detail in mind, it is important to notice that a higher rebalancing frequency leads to more injections of capital if the adjusting to risk exposure demands it. In this case, this is true, what can been seen in figure: 12
- Transaction Costs: transaction costs in fact can be overlooked since the higher frequency rebalancing strategy, that take more transaction costs, only presents in total $\approx 6,77$ €of transaction costs. This only represents $0,0677\%$ of the initial value invested. All the values can be seen in Fig. 9

As seen, the frequencies of 21 and 60 achieved better performance than 252 and 126. However, bearing in mind that higher frequencies does represent higher capital injection in increasing markets, we present the two construction methods that make use of CR (NRP and VT) with 21 and 60 day periods

of rebalancing, for those willing to invest the necessary capital that the strategy needs, and also the 252 period to those who are more conservative and don't have the availability to deposit frequently. (252 is preferred over 126 because it shown better returning).

D. Drift Rebalancing

The drift threshold bands chosen were 0.1, 0.05 and 0.01. The plot presented refers to the cumulative returns of the investment, comparing drift rebalancing with the buy and hold strategy.

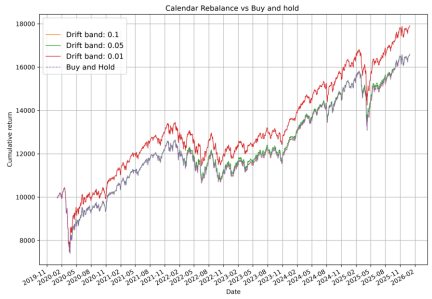


FIG. 2: Cumulative Ret. for different band thresholds - DR

Band	Inv. (€)	Final (€)	Annual (%)	Abs. Ret. (%)	Exc. Ret. (%)
10%	9,995.00	16,428.34	8.64	64.37	+0.62
5%	9,995.00	16,456.38	8.68	64.65	+0.66
1%	9,995.00	17,893.52	10.19	79.02	+0.82

TABLE VII: Drift Rebalancing — Comparison of strategies

Analysing Fig. 2 and Tab. VII:

- Rebalancing Advantage: it is obvious that the threshold of 0.01 ensured that the portfolio obtained higher cumulative returns than the other strategies, surpassing BH cumulative returs in $\approx 10\%$. The other bands achieved similar excess returns between them, surpassing the BH absolute return by $\approx 8\%$.
- Threshold band: thus, one can infer that the threshold of 0.05 and 0.1 made the trigger more difficult to action when compared to the 0.01 trigger, in accord with general portfolios constructed with ETFs risk-profile.
- Capital Injection: same as in CR. In this strategy, the rebalancing was more frequent in DR of 0.01 (Fig. 9).
- Transaction Costs: same as in CR. In this case the superior cost was about 5,80 (euros) (Fig. 9) of transactions cost, that only represents $0,058\%$ of the initial value invested.

E. Naive Risk Parity (NRP)

NRP method took advantage of the results of CR and did rebalancing frequencies of 21, 60 and 252 days, , with volatility look-back (VL) of 21 and 60 days. We first present the ex-ante expected measures of risk, so we can evaluate if the prediction of the portfolio was good.

Ex-Ante:

We divide the prediction in 2 parts: the first corresponds to the assessment of how big the incurred loss can be (using VaR and CVaR metrics) and then give the on-going ex-ante analysis, made by assessing the next day $(t+1)$ using the portfolio's allocation on day t.

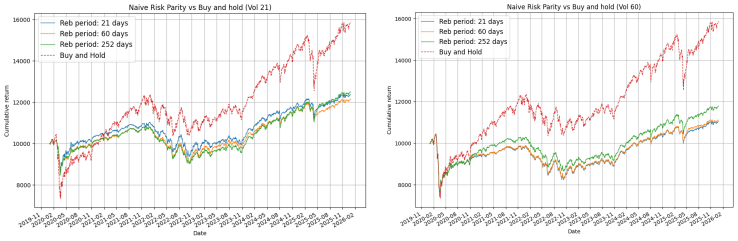
Reb.	VL	VaR (€)	$CVaR$ (€)	β	σ_p
21	21	323.46	610.09	0.501	0.0186
60	21	322.82	718.69	0.506	0.0184
252	21	322.82	718.69	0.517	0.0187
21	60	453.49	1661.81	0.723	0.0256
60	60	456.23	1611.23	0.739	0.0259
252	60	454.91	1582.68	0.746	0.0259

TABLE VIII: Ex-ante risk and performance metrics for NRP. VaR and CVaR represent the projected loss for the period 01-01-2025 to 31-12-2025; β is computed with ex-ante returns and σ_p are estimated over 01-01-2020 to 31-12-2024.

Analysing table VIII:

- The largest possible loss that can be incurred, given a 95% confidence level (VaR) is lower in the smaller case of VL (21 days) and does not varies much for different rebalancing periods.
- The expected loss in extreme circumstances, i.e. the expected value of losses pass the 5% confidence threshold imposed by VaR, for a VL of 60 days surpasses in more that an half every respective in rebalancing for the 21 period look-back. Interesting enough to be noticed is the inverse growing order behaviour between the different vol. look-back.
- Ex-ante β was significantly far from 1, ranging from 0.501 to 0.746. Low β demonstrates low systematic exposure, reducing to approximately one half and one quarter, respectively. It predicts that this strategy will systematically de-lever the portfolio, being extreme in lower VL, reducing the risk-exposure and thus not accompanying the volatility of the market.
- An evaluation of the ex-ante σ_p reveals minimal variance across both strategies. This low predictive metric anticipates that realized returns will exhibit low dispersion around the mean, validating the model's capacity to enforce a stable, constant risk-exposure profile over time, regardless of underlying market regimes.

From this evaluation we can predict that the risk-exposure will constantly de-lever, trying to reduce extreme losses but also conditioning gain. So, we now present 2 graphics, referring to the cumulative returns of this strategy and BH (in red) for reference, with different VLs: 21 and 60 days.



(a) Vol. look-back of 21 days (b) Vol. look-back of 60 days

FIG. 3: NRP for 3 rebalancing periods, 21, 60 and 252, and with different VLs

Reb.	VL	ER (%)	σ_{er}	IR	α	β	σ_p	MDD (%)
21	21	-4.851	0.0810	-9.505	≈ 0	0.467	0.0729	-15.929
60	21	-4.951	0.0815	-9.635	≈ 0	0.460	0.0717	-16.927
252	21	-4.389	0.0825	-8.446	≈ 0	0.457	0.0719	-16.927
21	60	-6.700	0.0688	-15.454	≈ 0	0.602	0.0929	-29.087
60	60	-6.139	0.0680	-14.326	≈ 0	0.605	0.0932	-29.087
252	60	-5.363	0.0640	-13.330	≈ 0	0.630	0.0952	-29.087

TABLE IX: Ex-post performance and risk metrics for NRP. ER, σ_{er}, IR, α and β are computed relative to the BH portfolio; σ_p and MDD describe the strategy itself.

Analysing Fig.3 and table IX:

- Strategy: NRP portfolios underperformed the BH portfolio in all of the cases, confirmed by analysing the ER (Excess Returns) that were consistently negative.
- Since rebalancing weights are inversely proportional to volatility, in high volatility markets the portfolio exposure will always be decreased by this strategy (comparing NRP portfolios' volatility with BH portfolio volatility - Tab. XVII in Appendix D).
- The volatility of ER was low for every case. Combined with the fact that the mean of ER was negative and with the visualisation of the plots, it can be deduced that the excess returns remained negatively stable. Exceptionally, that was not the case for the first years in Fig. 3a and months in Fig. 3b.
- VL of 21 days: the MDds do not differ much across rebalancing frequencies but are less accentuated than MDd of the BH portfolio (-29.1%). This means that smaller look-back periods reduce risk-exposure, providing a volatility estimate that prevents the same peak drawdown.
- VL of 60 days: the MDds are all equal across rebalancing frequencies and coincide with the MDd of the BH portfolio (-29.1%). This means that longer look-back periods provide a smoother volatility estimate that does not react quickly enough to prevent the same peak drawdown.
- IR was substantially negative, indicating that investment in this strategy did not compensate the costs associated with its active management profile. Incorporating an extended historical window of returns for volatility estimation adversely affects the performance of the strategy, as evidenced by the comparison between the two VL horizons. This outcome is consistent with the design of the strategy, which penalizes periods of elevated volatility; consequently, the additional drag introduced by including past high-volatility episodes can diminish overall returns, as also seen in ER, where the look-back of 60 days does give worse excess returns.
- α values appear to range near zero. This means that NRP generates no risk-adjusted outperformance — the underperformance compared to BH is entirely explained by the lower β, not by the performance of the underlying ETFs neither because of the period evaluated.
- β was significantly far from 1, ranging from $0,467$ to $0,630$. Low β demonstrates lower systematic exposure, reducing it in to approximately one half. Thus, it implies lower expected returns in a increasing market, explaining the negative ER results, despite lower volatility.

F. Volatility Targeting (VT)

VT method followed the same rebalancing frequencies as NRP, which were 21, 60 and 252 days, with VL of 21 and 60 days.

We first present the ex-ante expected measures of risk, so we can evaluate if the prediction of the portfolio is good.

Ex-Ante:

We divide the prediction in 2 parts: the first corresponds to the assessment of how big the incurred loss can be (using VaR and CVaR metrics) and then give the on-going ex-ante analysis, made by assessing the next day $(t+1)$ using the portfolio's allocation on day t.

Analysing table X:

- The largest possible loss that can be incurred, given a 95% confidence level (VaR), is lower in the smaller case of VL (21 days). Then, it deviates from other reb., whose values are similar to a VL of 60. The most extreme result (reb=60 and VL = 60) represents a loss of 6.15% on the initial investment of portfolio.
- CVaR under a 60-day VL exceeds the 21-day counterpart by more than 50%. Notably, the two horizons exhibit an inverse relationship: the 21-day window adapts rapidly to immediate market shifts, whereas the 60-day window reacts with a lag, capturing higher-magnitude tail risk.

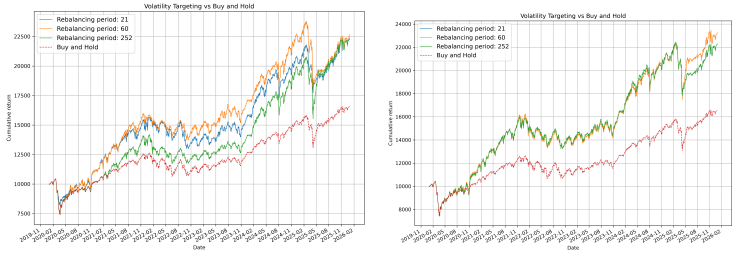
Reb.	Vol.	VaR (€)	$CVaR$ (€)	β	σ_p
21	21	525.10	787.00	0.850	0.0309
60	21	611.06	1382.83	1.033	0.0353
252	21	579.88	1450.79	1.041	0.0348
21	60	607.58	1444.28	1.062	0.0359
60	60	614.99	1443.29	1.094	0.0369
252	60	589.92	1443.29	1.062	0.0359

TABLE X: Ex-ante risk and performance metrics for VT. VaR and CVaR represent the projected loss for the period 01-01-2025 to 31-12-2025; β is computed with ex-ante returns and σ_p are estimated over 01-01-2020 to 31-12-2024.

- Except for the initial configuration, the ex-ante $\beta > 1$ across most scenarios, indicating that the framework systematically amplifies benchmark market exposure. This structural leverage is a direct consequence of the volatility-scaling mechanism during periods of low market variance.
- Ex-ante σ_p: The ex-ante portfolio volatility (σ_p) remains low and tight across both VL horizons. This highly compressed risk profile anticipates bounded return dispersion and suggests that varying the look-back window length does not fundamentally alter the portfolio's structural risk-exposure.

From this evaluation, we can predict that the risk-exposure will be following the same tendency as the BH portfolio, so the same as the market.

We, again, present 2 plots referring to the cumulative ex-post returns of the strategy and BH for reference, with different VLs.



(a) Vol. look-back of 21 days (b) Vol. look-back of 60 days

FIG. 4: VT for 3 rebalancing periods, 21, 60 and 252, and with different VLs

Reb.	Vol.	ER (%)	σ_{er}	IR	α	β	σ_p	MDd (%)
21	21	5.842	0.0673	0.868	0.000210	1.086	0.1643	-21.603
60	21	6.420	0.0622	1.033	0.000202	1.203	0.1755	-29.169
252	21	6.585	0.0510	1.291	0.000214	1.180	0.1693	-29.169
21	60	5.859	0.0554	1.058	0.000184	1.185	0.1713	-29.099
60	60	6.750	0.0562	1.201	0.000213	1.211	0.1744	-29.099
252	60	5.859	0.0554	1.058	0.000184	1.185	0.1713	-29.099

TABLE XI: Ex-post performance and risk metrics for VT. ER, σ_{er} , IR, α and β are computed relative to the BH portfolio; σ_p and MDd describe the strategy itself

Analysing Fig.4 and table XI:

- Excess returns are positive across all rebalancing frequencies, meaning that VT portfolios outperformed the BH portfolio in all cases. The leverage mechanism allows the strategy to amplify exposure in low-volatility regimes, capturing additional upside.
- The volatility of excess returns was low for every case. Moreover, target weights are scaled to meet a volatility target, so the strategy aims to actively increase exposure in calm markets and reduces it in turbulent ones.

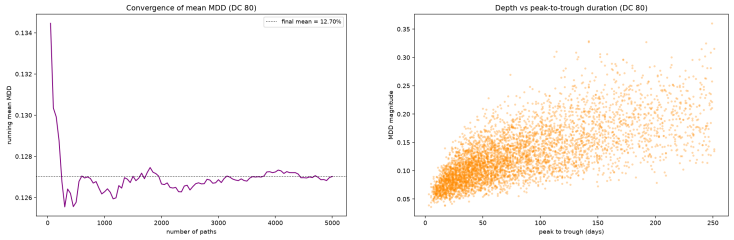
- For the case of VL of 60 days, the MDd values are analogous to the NRP case. However, for 21 days, the MDD for reb=21 is lower (-21.6%) than for reb=60 and reb=252 (-29.2%), indicating that a higher rebalancing frequency offers better drawdown protection.
- IR was positive in all cases, showing that investing in this strategy compensates the risk of the transaction costs inherent to active management. The pattern is visible: for vol=21, IR increases with rebalancing period; for vol=60 it's less monotone.
- $\beta > 1$, meaning that the VT portfolio amplifies the market exposure of BH. This is a direct consequence of leverage, as in the low-volatility regimes the strategy scales up allocation beyond the target weights, increasing systematic exposure above that of a fully-invested portfolio.
- α ranges near zero, meaning that in fact excess returns can be completely explained by β.

G. Monte-Carlo analysis

Now, we turn to a forward-looking assessment based on a Monte-Carlo simulation for a BH portfolio. Each asset price is modelled as a Geometric Brownian Motion, with per-asset drift μ_i and the covariance matrix Σ estimated from the historical daily log-returns over the estimation window (01-01-2020 to 31-12-2024). We apply the Cholesky decomposition of Σ, so that the movement between correlated assets is preserved rather than each asset being drawn independently.

We simulate $N = 5000$ paths over a one-year period (252 trading days). The portfolio is held as buy-and-hold, the units are fixed at $t = 0$ from the static target weights and the NAV evolves as $\mathrm{NAV}_t = V_0 \sum_i w_i G_{i,t}$, where $G_{i,t}$ is the simulated gross growth of asset i.

It is also worth stressing what this model does and does not provide. The drift is estimated from a historical average and is held constant, so the simulation does not forecast the future direction of prices. What it actually does is produce a distribution of plausible outcomes under the assumption that the return statistics remain stationary. As a consequence, the object of interest is the risk profile of the portfolio and the Expected Maximum Drawdown (EMDd).



(a) Convergence of the mean MDD estimate with the number of paths.

(b) Drawdown depth against its peak-to-trough duration.

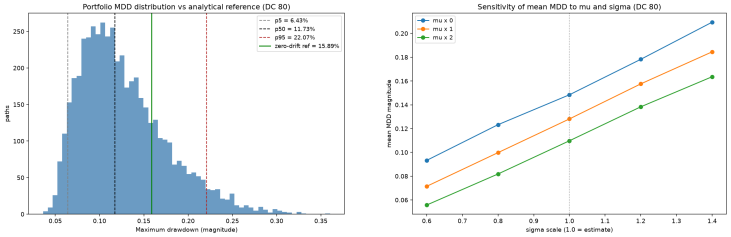
FIG. 5: Monte-Carlo diagnostics for the DC 80 portfolio: estimator convergence and the depth-duration relationship.

By noticing that mean MDd is itself a Monte-Carlo average, it makes sense to check for convergence before any value is reported. As a result, we start by analysing figure 5a, which shows the running mean of the absolute MDd as paths are accumulated. After an initial high-variance transient below roughly 500 paths, the estimate starts stabilising around the final mean, 12.70%, and remains flat from approximately 2000 to 3000 paths onward. We also check that $N = 5000$ is enough as the residual sampling noise is negligible and the reported statistics should be stable.

Figure 5b relates the depth of each path's maximum drawdown to the number of days elapsed from its peak to its trough. We can observe that deeper drawdowns tend to take longer to develop. The bulk of the paths cluster at short durations (under 50 days) and shallow depths (around 10 days), whereas the most severe drawdowns (above 25%) occur mostly, if not exclusively, at long durations. We conclude that, under the GBM dynamics, large drawdowns are not abrupt one-day events but accumulate gradually over time.

Figure 6a shows the full distribution of the portfolio maximum drawdown across the 5000 paths. We observe that the median is 11.73%, the 5th to 95th percentile range is 6.43% to 22.07% and a thin tail extends beyond 30%. The analytical zero-drift is computed using Ref. [15], shown as the vertical line at 15.89%. It lies above the simulated median because it assumes no drift. The positive estimated drift of the portfolio decreases the drawdowns and shifts the distribution to the left. When we have such a skewed plot, reporting the median together with the 95th percentile is more informative than the mean, so this analysis was necessary.

Then, figure 6b reports the mean MDD as the volatility and drift inputs are rescaled, each line corresponding to a different drift multiplier. We can observe the dependence on σ is strong and approximately linear. The drift has an opposite effect as for $\mu \times 2$, the curve lies below $\mu \times 1$ and $\mu \times 0$ (drift-free) curve. The practical implication is that the EMDd is governed primarily by the volatility estimate, which means that an estimation error in σ is the dominant source of uncertainty in the forecast. We also check that the notoriously noisy drift estimate matters little in this context.



(a) Distribution of the portfolio maximum drawdown.

(b) Sensitivity of the mean MDD to the μ and σ inputs.

FIG. 6: Monte-Carlo drawdown distribution and the sensitivity of the Expected Maximum Drawdown to the estimated inputs.

Figure 7 compares the portfolio EMDd with that of each individual ETF. The portfolio mean MDD (12.7%) is lower than that of the two equity ETFs, C (14.8%) and A (16.9%), while B is the least risky of the three (6.2%). Remember that ETF B invests in bonds whereas the others belong in the equity class. Thus, B is therefore less riskier than A or C. This is a great way to check that the simulation is properly working.

Additionally, the portfolio EMDd is also lower than the weight-averaged EMDd of its constituents because the assets are not perfectly correlated. So, by combining them, we reduce the drawdown risk below what the individual exposures would suggest. This is exactly the diversification benefit that we aim with this approach and it is captured here only because the simulation preserves the asset correlations through the Cholesky step. As in the distribution above, the zero-drift reference exceeds the simulated mean for every series, reflecting the cushioning effect of the positive drift.

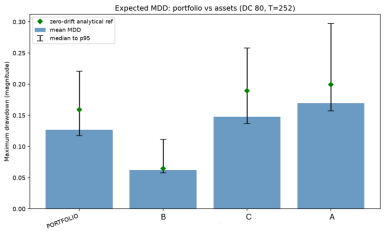


FIG. 7: Expected maximum drawdown of the portfolio compared with each constituent ETF. Bars show the mean MDD, whiskers the median to 95th percentile range, and the markers the zero-drift analytical reference.

Finally, figure 8 confronts the model with reality. We estimate using data from 2020 until 2024 and simulate the 2025 calendar year. We overlay the realised buy-and-hold NAV on the simulated percentile fan. Two observations stand out. First, the last NAVv value (approximately $11,027$) lands close to the simulated median, so the final outcome was well within the model's central expectation. Second, the path taken does not account real world events. On 2 April, the USA announced the new tariff policy (dashed line), which resulted in the portfolio falling sharply and breached the simulated 5th-percentile NAV envelope around its trough, before recovering over the remainder of the year. This abrupt, event-driven move is precisely the kind of discontinuity that this type of model cannot reproduce and it illustrates the well-known tendency of such models to understate tail risk.

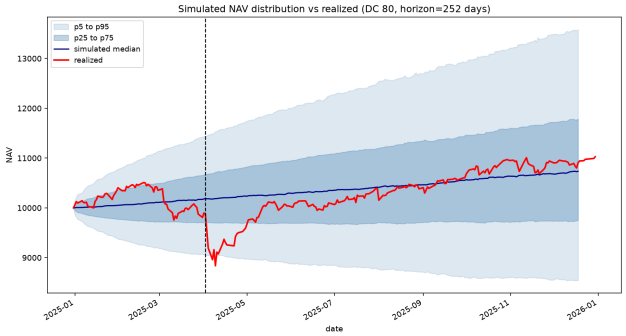


FIG. 8: Out-of-sample validation: the simulated NAV distribution (percentile fan, estimated on 2020 to 2024) against the realized buy-and-hold path over 2025 (red). The dashed line marks the 2nd of April tariff announcement.

In addition to it, the maximum realised drawdown (15.91%) still fell within the simulated 5 to 95th percentile band (6.43% to 22.07%). The magnitude of the worst drawdown remained plausible under the model even though its timing did not. Overall, we can say that the simulation captures the central tendency and the plausible range of the portfolio well, but not the jump dynamics of an exogenous shock, like a Trump presidency.

VIII. Conclusions

The quantative filter mechanism for the evaluation of replication, cost and liquidity successfully elected the 4 analysed funds. Note that those were benchmarked against other ETFs that respectively tracked the same index. Hence, the scoring system analysed how different ETFs that track the same index behave in relation to one another. We found that for ETF B the replication was very different, indicting that they do not behave the same.

For rebalancing strategies we found that both CR and DR outperformed a static Buy and Hold portfolio, showed by the excess in the cumulative returns. This demonstrates that periodically reseting a portoflio to capture market fluctuations adds clear value. Highlighting the conditions imposed in this strategies, we found that for CR, shorter intervals yielded higher returns with the monthly and quarterly periods performing the best. For DR, a tight 1% deviation band forced more frequent adjustments and generated the higher excess returns, surpassing the baseline BH portfolio for approximately 8%. To meet strategy needs, the transaction costs were notably negligible, since that even fore more frequent rebalancing it accounted for less of 0.07% of the initial capital invested.

The next two presented portfolio construction strategies, took advantage of the BH portfolio performance results. Under that framework, NRP consistently shown underperformance under increasing markets when compared to BH, producing negative excess returns. Since weights are inversely proportional to volatility, this strategy systematically de-levers the portfolio during highly volatile periods, causing it to miss out on increasing market gains. Despite lower returns, NRP diminished risk-exposure. The shorter 21-day VL window allowed the portfolio to respond quickly to huge shifts on the markets, protecting the portfolio from massive drawdowns that can incur massive losses. The same advantage is not notorious when looking to the performance with quarterly information on volatility. In contrast with NRP, VT approach outperformed BH across the different impositions: by using a volatility-scaling mechanism, the strategy dynamically leverages the portfolio ($\beta > 1$) during low-volatility regimes. This active strategy amplifies market exposure to ensure that during calm markets the gains can be maximized. Nevertheless, similar to NRP, higher rebalancing frequencies provided superior drawdown protection over longer rebalancing intervals. The forward-looking Monte Carlo simulation validated the structural value of diversification. By incorporating a Cholesky decomposition to map asset conjoint evolution, the simulation accurately showed that EMDd was lower than its individual equity components and lower than a basic weighted average. When validated against real historical data from the 2025 year, the simulated portfolio tendency median folowed close the model's central media expectation. However, the model failed to replicate abrupt and event-driven schocks, such as the sharp portfolio drop following a U.S. tariff policy announcement on April 2nd. This highlights the fact that continuous-time stochastic models, like GBM, still fails to understand rapidly switching dynamics.

References

[1] H. Markowitz, Portfolio selection, The Journal of Finance **7**, 77 (1952).

[2] J. M. Hill, D. Nadig, and M. Hougan, *A Comprehensive Guide to Exchange-Traded Funds (ETFs)* (CFA Institute Research Foundation, 2015) with an appendix on international ETFs by Deborah Fuhr.

[3] P. A. Samuelson, Rational theory of warrant pricing, Industrial Management Review **6**, 13 (1965).

[4] European Parliament and Council of the European Union, Regulation (eu) no 575/2013 on prudential requirements for credit institutions and investment firms, article 339(2)(b), Official Journal of the European Union (2013).

[5] Y. Zilbering, C. M. Jaconetti, and F. M. Kinniry Jr., *Best practices for portfolio rebalancing*, White Paper (Vanguard Research, 2015).

[6] E. E. Qian, *Risk Parity Portfolios: Efficient Portfolios Through True Diversification*, White Paper (PanAgora Asset Management, 2005).

[7] S. Maillard, T. Roncalli, and J. Teïletche, The properties of equally weighted risk contribution portfolios, The Journal of Portfolio Management **36**, 60 (2010).

[8] QuantInsti, Risk parity portfolio: Strategy, example & python implementation, QuantInsti Blog (2025).

[9] K. Dachraoui, On the optimality of target volatility strategies, The Journal of Portfolio Management **44**, 58 (2018).

[10] C. Frey and B. Uhl, Tomorrow never dies: The impact of measuring portfolio risk ex-ante and ex-post (2023), draft version. Christoph Frey: Lancaster University and Pinechip Capital GmbH; Björn Uhl: Pinechip Capital GmbH.

[11] T. Hamacher, *Measuring the Investment Risk of Equity Funds: A Reference for Fund Directors*, Reference Guide (Mutual Fund Directors Forum, 2018) reviewed by Dave Butler and the risk team at Calamos Investments.

[12] F. Stambaugh, Risk and value at risk, European Management Journal **14**, 612 (1996).

[13] MIT OpenCourseWare, Lecture 5: Stochastic processes i, 18.S096 Topics in Mathematics with Applications in Finance (2013), fall 2013. MIT OpenCourseWare.

[14] Y. Saoudi, K. Jeaab, and H. Hachimi, Pricing basket options using Monte Carlo simulation employing Cholesky decomposition and variance reduction techniques under the 2D stochastic Black–Scholes equation, Partial Differential Equations in Applied Mathematics **15**, 101270 (2025).

[15] M. Magdon-Ismail, A. F. Atiya, A. Pratap, and Y. S. Abu-Mostafa, On the maximum drawdown of a brownian motion, Journal of Applied Probability **41**, 147 (2004)

A. ETF Filters

Metric	Limit type	Value
Correlation	minimum	0.95
Beta	minimum and maximum	0.8 and 1.2
TE	maximum	0.01
TER	maximum	0.005
ADTV	minimum	1×10^6
Spread (%)	maximum	0.5

TABLE XII: Eligibility thresholds per metric.

B. Scoring Systems

For metrics where a larger value is better (monotone increasing):

$$\text{score}(x) = \begin{cases} 0, & x \le \text{floor} \\ \dfrac{x - \text{floor}}{\text{cap} - \text{floor}}, & \text{floor} < x < \text{cap} \\ 1, & x \ge \text{cap} \end{cases} \tag{B1}$$

For metrics where a smaller value is better (monotone decreasing):

$$\text{score}(x) = \begin{cases} 1, & x \le \text{floor} \\ \dfrac{\text{cap} - x}{\text{cap} - \text{floor}}, & \text{floor} < x < \text{cap} \\ 0, & x \ge \text{cap} \end{cases} \tag{B2}$$

And lastly, for metrics that are best at a target value, with $d = |x - \text{target}|$:

$$\text{score}(x) = \begin{cases} 1, & d \le \text{tol} \\ \dfrac{\text{max_dev} - d}{\text{max_dev} - \text{tol}}, & \text{tol} < d < \text{max_dev} \\ 0, & d \ge \text{max_dev} \end{cases} \tag{B3}$$

C. Buy & Hold until Trigger condition

Method	Trigger condition
NRP	defined period of time - activated on day t
VT	defined period of time - activated on day t
CR	defined period of time - activated on day t
DR	deviation of the target weights

TABLE XIII: Methods and its trigger conditions

D. Buy and Hold Portfolio metrics

Init. Inv. (€)	Final Value (€)	Annual Ret. (%)	Absolute Ret. (%)
9,995.00	15,868.60	8.02	58.77

TABLE XIV: Buy and Hold returns

VaR (€)	$CVaR$ (€)
537.34	1438.93

TABLE XV: Possible loss incurred for the period of 01-01-2025 to 31-12-2025

β	σ_p
1.0	0.0321

TABLE XVI: Ex-ante Performance and Risk assessment metrics for the period of 01-01-2020 to 30-12-2025

σ_p	MDD (%)
0.1384	-29.744

TABLE XVII: Metrics about the portfolio strategy itself

E. Rebalance Costs and Capital Injection

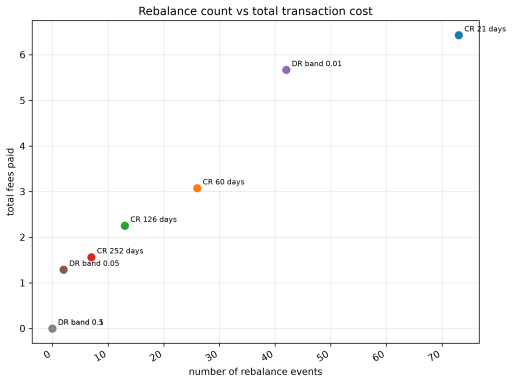


FIG. 9: Rebalance Costs Calendar and Drift Rebalancing

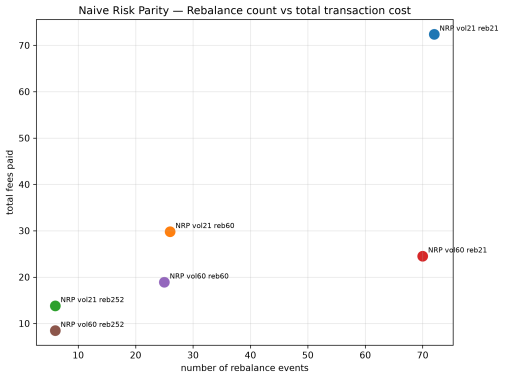


FIG. 10: Rebalance Costs Naive Risk Parity

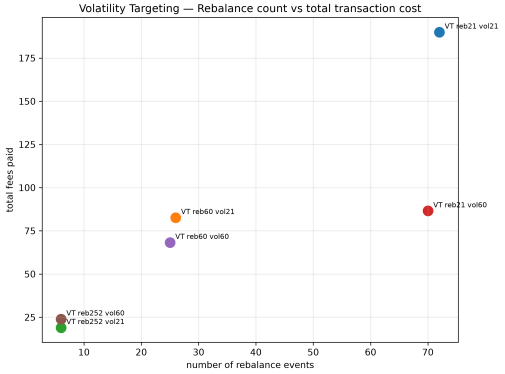


FIG. 11: Rebalance Costs Volatility Targeting

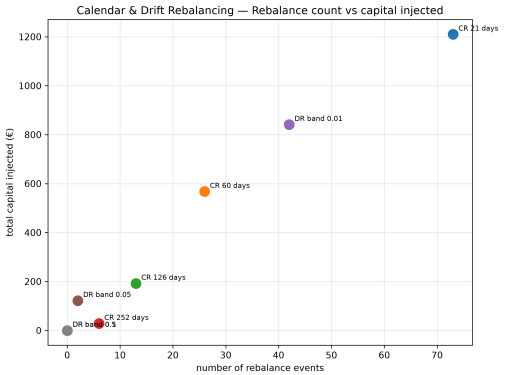


FIG. 12: Capital Injected for Calendar and Drift Rebalancing

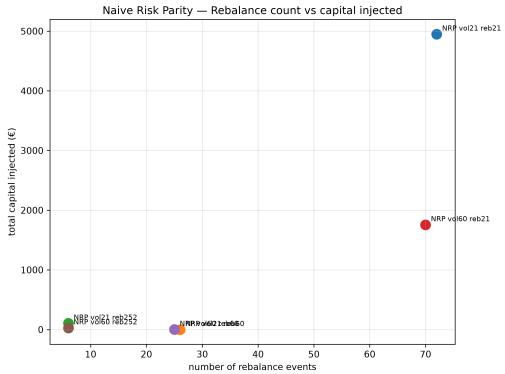


FIG. 13: Capital Injected and Naive Risk Parity



FIG. 14: Capital Injected and Volatility Targeting

F. Geometric Brownian Motion under Stochastic Calculus

Asset price dynamics are assumed to follow a GBM,governed by the stochastic differential equation (SDE):

$$dS(t) = \mu S(t)\,dt + \sigma S(t)\,dW(t) \tag{F1}$$

Solution: let $f(S(t)) = \ln(S(t))$
Applying Itô's Formula to $d(\ln(S(t))) = d\,f(S(t))$:

$$\begin{aligned} d(\ln S(t)) &= \frac{\partial f}{\partial x}\,dS(t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\,(dS(t))^2 \\ &= \frac{1}{S(t)}\,\mu S(t)\,dt + \frac{1}{S(t)}\,\sigma S(t)\,dW(t) - \qquad \text{(F2)} \\ &\quad - \frac{1}{2}\frac{1}{(S(t))^2}\,\sigma^2 (S(t))^2\,dt \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW(t) \qquad \text{(F3)} \end{aligned}$$

Integrating from t to τ:

$$\ln(X(\tau)) = \ln(S(t)) + \left(\mu - \frac{1}{2}\sigma^2\right)(\tau - t) + \sigma\,[W(\tau) - W(t)] \tag{F4}$$

$$X(\tau) = S(t)\,\exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)(\tau - t) + \sigma\,[W(\tau) - W(t)]\right] \tag{F5}$$